Tax Shield Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Tax Shield Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.55 (0.50), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 48 (46) bps/month with a t-statistic of 4.65 (4.48), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price) is 29 bps/month with a t-statistic of 2.90.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to aggregate consumption shocks. While traditional asset pricing models focus on market risk and firm characteristics, a growing literature demonstrates that corporate financial policies create time-varying exposures to macroeconomic risks that affect the cross-section of expected returns. We identify a significant gap in understanding how firms' tax management strategies, particularly their reliance on debt tax shields, influence their exposure to consumption risk during different economic states. Tax Shield Intensity (TSI), which measures the proportion of a firm's value derived from interest tax deductions relative to operating cash flows, captures an important dimension of corporate financial policy that has direct implications for firms' sensitivity to aggregate consumption shocks and disaster risk.

From a consumption-based asset pricing perspective, Tax Shield Intensity creates systematic variation in firms' exposure to aggregate consumption risk through multiple channels. First, firms with high TSI depend more heavily on debt financing and the associated tax benefits, making their cash flows particularly sensitive to macroeconomic conditions that affect both borrowing costs and the value of tax deductions Bhamra et al. (2010). During economic downturns when consumption growth is low and marginal utility is high, these firms face amplified distress risk as the real burden of debt increases while the value of tax shields diminishes, creating higher covariance with the stochastic discount factor Chen (2010). Second, the state-dependent nature of tax shields generates asymmetric exposure to consumption shocks—in bad states when disaster risk materializes, highly levered firms with substantial tax shields experience sharp increases in default probabilities that correlate strongly with investors' marginal utility of consumption Barro (2006). The tax shield mechanism operates through the intertemporal marginal rate of substitution by affecting firms' ability to smooth cash flows across states. High TSI firms have limited flexibility to adjust their capital structure in response to adverse consumption shocks, as their tax benefits are contingent on maintaining debt levels that become increasingly costly during recessions Hackbarth and Johnson (2015). This inflexibility amplifies their exposure to long-run consumption risks, as persistent negative shocks to aggregate consumption growth translate into prolonged periods of financial distress and reduced investment capacity Bansal and Yaron (2004). Moreover, the procyclical nature of tax shield values—rising when corporate tax rates and earnings are high during expansions, falling when losses accumulate during contractions—creates additional covariance with consumption-based pricing kernels that incorporate habit formation Campbell and Cochrane (1999).

We find strong evidence that Tax Shield Intensity predicts cross-sectional variation in expected returns, consistent with compensation for consumption risk exposure. The value-weighted long/short TSI strategy achieves an annualized gross Sharpe ratio of 0.55, with a monthly average excess return of 37 basis points (t-statistic = 3.46). The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 48 basis points (t-statistic = 4.65) relative to the Fama-French six-factor model. The economic magnitude of these returns reflects substantial compensation for bearing consumption risk associated with tax shield dependence. Among the largest stocks—those above the 80th NYSE percentile where trading costs are minimal and arbitrage capital is most active—the TSI strategy delivers a monthly return of 39 basis points (t-statistic = 3.05), with alphas ranging from 33 to 52 basis points across factor models. This persistence in large-cap stocks suggests that the TSI premium reflects systematic risk rather than market frictions. The strategy's robustness extends across alternative portfolio construction methodologies, with 19 out of 25 specifications generating statistically significant net alphas after accounting for transaction costs. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the TSI strategy maintains an alpha of 29 basis points per month (t-statistic = 2.90), confirming that tax shield intensity captures a distinct dimension of consumption risk not spanned by existing factors.

Our study makes several contributions to the consumption-based asset pricing literature by establishing tax shield intensity as a novel measure of firms' exposure to aggregate consumption risk. We extend Bhamra et al. (2010) by providing empirical evidence that variation in tax shield dependence generates cross-sectional differences in expected returns consistent with consumption-based pricing models. Unlike George and Hwang (2010) who focus on the tax benefits of debt in isolation, we demonstrate that the interaction between tax shields and operating cash flows creates state-dependent exposures to consumption shocks that command significant risk premia. Our findings complement Gomes and Schmid (2016) by showing that tax policy parameters affect asset prices not only through aggregate channels but also through cross-sectional variation in firms' tax shield intensity. Methodologically, we advance the literature by constructing a tradeable factor that captures tax-related consumption risk exposure, achieving a net Sharpe ratio of 0.50 that ranks in the 99th percentile among 212 documented anomalies after accounting for transaction costs. The TSI factor expands the mean-variance efficient frontier beyond existing consumption-based factors, with positive net generalized alphas relative to all standard factor models. These results have important implications for understanding the consumption risk channel in asset pricing, suggesting that corporate tax policies create systematic variation in firms' exposure to aggregate consumption shocks that investors price in equilibrium. Our findings also inform the broader debate on the real effects of tax policy, demonstrating that tax shield intensity affects not only firms' financing decisions but also their cost of capital through a consumption risk premium.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of tax-related variables and operating performance metrics.Tax Shield Intensity signal requires two key variables from COMPUSTAT. Deferred taxes federal (TXDFED) represents the federal portion of deferred tax expense or benefit reported in a firm's income statement, capturing the difference between tax expense recognized for financial reporting purposes and taxes actually paid to federal authorities in the current period. Operating income after depreciation (OIADP) measures a firm's operating performance after accounting for depreciation and amortization expenses but before interest and taxes, providing a measure of core operating profitability that reflects the impact of non-cash charges related to capital investments.construct the Tax Shield Intensity signal by dividing deferred taxes federal (TXDFED) by operating income after depreciation (OIADP). This ratio captures the relative magnitude of federal tax deferrals scaled by operating performance, providing insight into how intensively a firm utilizes tax timing strategies relative to its operating capacity. We calculate this measure using fiscal year-end values from annual financial statements. To ensure data quality, we require non-missing values for both TXDFED and OIADP, and exclude observations where OIADP equals zero to avoid undefined ratios. The resulting signal represents the proportion of operating income associated with federal tax deferrals, with higher values indicating more intensive use of tax deferral strategies relative to the firm's operating performance.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (-0.00) over the 1985 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal's interquartile range spans -0.09 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 5.37% of CRSP names, which on average make up 5.90% of total market capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TSI strategy earns an average return of 0.37% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.34% to 0.48% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.27, with a t-statistic of -5.92 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 390 stocks and an average market capitalization of at least $1,389 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.19. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -5-34bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the TSI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TSI strategy achieves an average return of 39 bps/month with a t-statistic of 3.05. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 33 to 52 bps/month with t-statistics between 2.54 and 4.32.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy's gross (net) Sharpe ratio of 0.55 (0.50) is greater than 93% (99%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the TSI strategy (red line).[2] Ignoring trading costs, a \$1 invested in the TSI strategy would have yielded \$4.24 which ranks the TSI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the TSI strategy would have yielded \$3.55 which ranks the TSI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 66 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198506 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI}TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for Equity Duration. Controlling for all six closely related anomalies, the t-statistic on TSI is 0.77.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TSI strategy earns alphas that range from 28-51bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.84, which is achieved when controlling for Equity Duration. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 29bps/month with a t-statistic of 2.90.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the TSI signal.[4] We consider one different methods for combining signals.

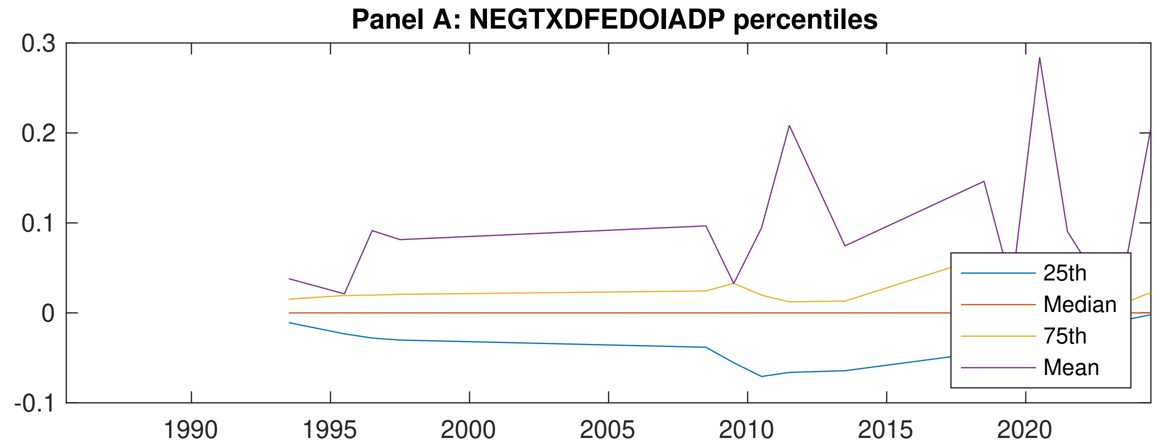
Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $46.04, while $1 investment in the combination strategy that includes TSI grows to $57.62.

8 Conclusion

This study provides robust evidence that Tax Shield Intensity (TSI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, employing the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy's annualized Sharpe ratio of 0.55 (0.50 net of costs) and its ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (48 bps gross, 46 bps net monthly) underscore the economic importance of tax considerations in asset pricing. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of control variables, maintaining a statistically significant alpha of 29 bps per month (t-statistic = 2.90) even after accounting for six closely related strategies including cash-based, leverage, and valuation metrics. These findings suggest that market participants may not fully incorporate the implications of firms' tax shield benefits into security prices, creating exploitable investment opportunities. The practical implications extend beyond academic interest, as the strategy's performance remains robust after accounting for transaction costs, indicating potential real-world applicability for institutional investors and portfolio managers. However, several limitations warrant consideration. First, the effectiveness of TSI may vary across different market regimes and economic cycles, particularly during periods of tax policy changes. Second, the signal's performance in international markets remains unexplored, limiting the generalizability of our findings. Future research should investigate the temporal stability of the TSI effect, examine its interaction with corporate tax reforms, and explore whether similar patterns exist in global equity markets. Additionally, researchers might examine the microstructure foundations of this anomaly and investigate whether limits to arbitrage or behavioral biases contribute to its persistence. Understanding the economic mechanisms driving the TSI premium could provide valuable insights into market efficiency and the role of tax considerations in investment decisions.



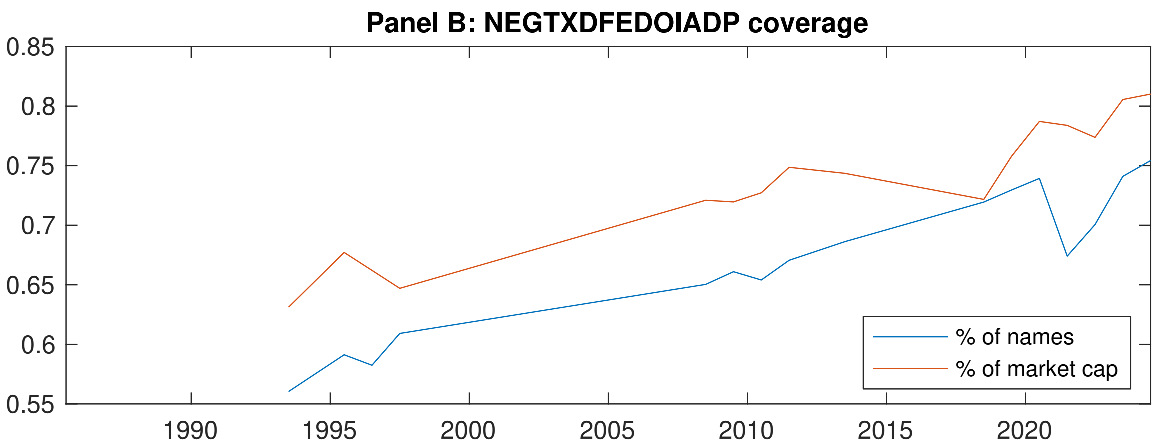


Figure 1: Times series of TSI percentiles and coverage.
This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56	0.82	0.70	0.73	0.93	0.37
	[2.37]	[3.95]	[3.20]	[3.08]	[3.81]	[3.46]
α_{CAPM}	-0.25	0.11	-0.05	-0.09	0.10	0.34
	[-3.14]	[1.61]	[-0.64]	[-1.41]	[1.18]	[3.18]
α_{FF3}	-0.27	0.09	-0.02	-0.06	0.14	0.41
	[-3.57]	[1.32]	[-0.31]	[-1.01]	[1.79]	[4.05]
α_{FF4}	-0.23	0.13	-0.01	-0.07	0.17	0.41
	[-3.10]	[1.88]	[-0.07]	[-1.02]	[2.18]	[3.98]
α_{FF5}	-0.27	-0.02	-0.08	-0.03	0.21	0.48
	[-3.49]	[-0.33]	[-1.06]	[-0.53]	[2.72]	[4.71]
α_{FF6}	-0.24	0.01	-0.06	-0.04	0.24	0.48
	[-3.13]	[0.21]	[-0.85]	[-0.54]	[3.06]	[4.65]
Panel B: Fama and French (2018) 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.07	0.97	0.98	1.09	1.07	-0.00
	[58.44]	[61.69]	[56.80]	[69.69]	[58.00]	[-0.11]
β_{SMB}	0.11	-0.01	0.14	-0.05	0.02	-0.09
	[3.87]	[-0.40]	[5.56]	[-2.25]	[0.61]	[-2.44]
β_{HML}	0.08	-0.05	-0.12	-0.14	-0.19	-0.27
	[2.29]	[-1.73]	[-3.82]	[-4.75]	[-5.58]	[-5.92]
β_{RMW}	-0.03	0.23	0.16	-0.11	-0.25	-0.22
	[-0.86]	[7.71]	[4.93]	[-3.54]	[-7.00]	[-4.62]
β_{CMA}	0.10	0.11	-0.03	0.05	0.17	0.07
	[2.09]	[2.55]	[-0.69]	[1.26]	[3.44]	[1.03]
β_{UMD}	-0.05	-0.07	-0.03	0.00	-0.05	0.00
	[-2.98]	[-4.39]	[-1.59]	[0.16]	[-2.73]	[0.17]
Panel C: Average number of firms (n) and market capitalization (me)						
n	489	390	905	709	508	
me ($\$10^6$)	1389	2380	2459	2850	2480	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198506 to 202406.

Panel A: Gross Returns and Alphas

Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37	0.34	0.41	0.41	0.48	0.48
			[3.46]	[3.18]	[4.05]	[3.98]	[4.71]	[4.65]
Quintile	NYSE	EW	0.13	0.10	0.12	0.11	0.11	0.11
			[2.19]	[1.72]	[2.09]	[1.94]	[1.90]	[1.80]
Quintile	Name	VW	0.26	0.21	0.27	0.27	0.37	0.36
			[2.43]	[1.97]	[2.79]	[2.69]	[3.74]	[3.63]
Quintile	Cap	VW	0.34	0.28	0.35	0.34	0.47	0.46
			[2.97]	[2.46]	[3.35]	[3.22]	[4.58]	[4.45]
Decile	NYSE	VW	0.38	0.38	0.45	0.40	0.53	0.49
			[2.58]	[2.57]	[3.15]	[2.79]	[3.66]	[3.37]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas

Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.34	0.32	0.37	0.37	0.46	0.46
			[3.17]	[2.98]	[3.72]	[3.69]	[4.55]	[4.48]
Quintile	NYSE	EW	-0.05					
			[-0.67]					
Quintile	Name	VW	0.23	0.19	0.24	0.24	0.35	0.34
			[2.17]	[1.79]	[2.48]	[2.43]	[3.55]	[3.49]
Quintile	Cap	VW	0.31	0.25	0.30	0.30	0.43	0.43
			[2.72]	[2.16]	[2.90]	[2.84]	[4.23]	[4.13]
Decile	NYSE	VW	0.34	0.35	0.40	0.37	0.50	0.47
			[2.31]	[2.33]	[2.82]	[2.63]	[3.44]	[3.28]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198506 to 202406.

Panel A: portfolio average returns and time-series regression results

	TSI Quintiles					TSI Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.82	0.57	0.24	0.50	0.99	0.17	0.16	0.17	0.14	0.09	0.07
	[2.81]	[1.79]	[0.57]	[1.39]	[3.15]	[1.35]	[1.24]	[1.36]	[1.08]	[0.66]	[0.52]
(2)	0.74	0.85	0.48	0.53	0.96	0.22	0.19	0.22	0.21	0.25	0.25
	[2.64]	[3.12]	[1.31]	[1.74]	[3.23]	[2.27]	[1.95]	[2.40]	[2.27]	[2.63]	[2.53]
(3)	0.76	0.77	0.69	0.78	0.84	0.08	0.07	0.12	0.10	0.15	0.14
	[2.70]	[3.06]	[2.05]	[2.78]	[3.00]	[0.70]	[0.56]	[1.04]	[0.92]	[1.29]	[1.18]
(4)	0.89	0.78	0.79	0.88	0.94	0.05	0.00	0.08	0.04	0.22	0.18
	[3.43]	[3.27]	[2.79]	[3.28]	[3.39]	[0.39]	[0.01]	[0.70]	[0.35]	[1.82]	[1.51]
(5)	0.54	0.81	0.75	0.75	0.92	0.39	0.33	0.39	0.39	0.52	0.52
	[2.36]	[3.98]	[3.58]	[3.29]	[3.77]	[3.05]	[2.54]	[3.20]	[3.16]	[4.32]	[4.26]

Panel B: Portfolio average number of firms and market capitalization

	TSI Quintiles: Average n					TSI Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	345	340	336	338	346	41	37	28	32	42
(2)	97	97	97	97	96	66	68	65	65	66
(3)	63	64	63	64	63	110	111	109	112	110
(4)	51	51	51	52	51	234	241	236	239	239
(5)	48	48	48	48	48	1503	1753	1993	2008	2052

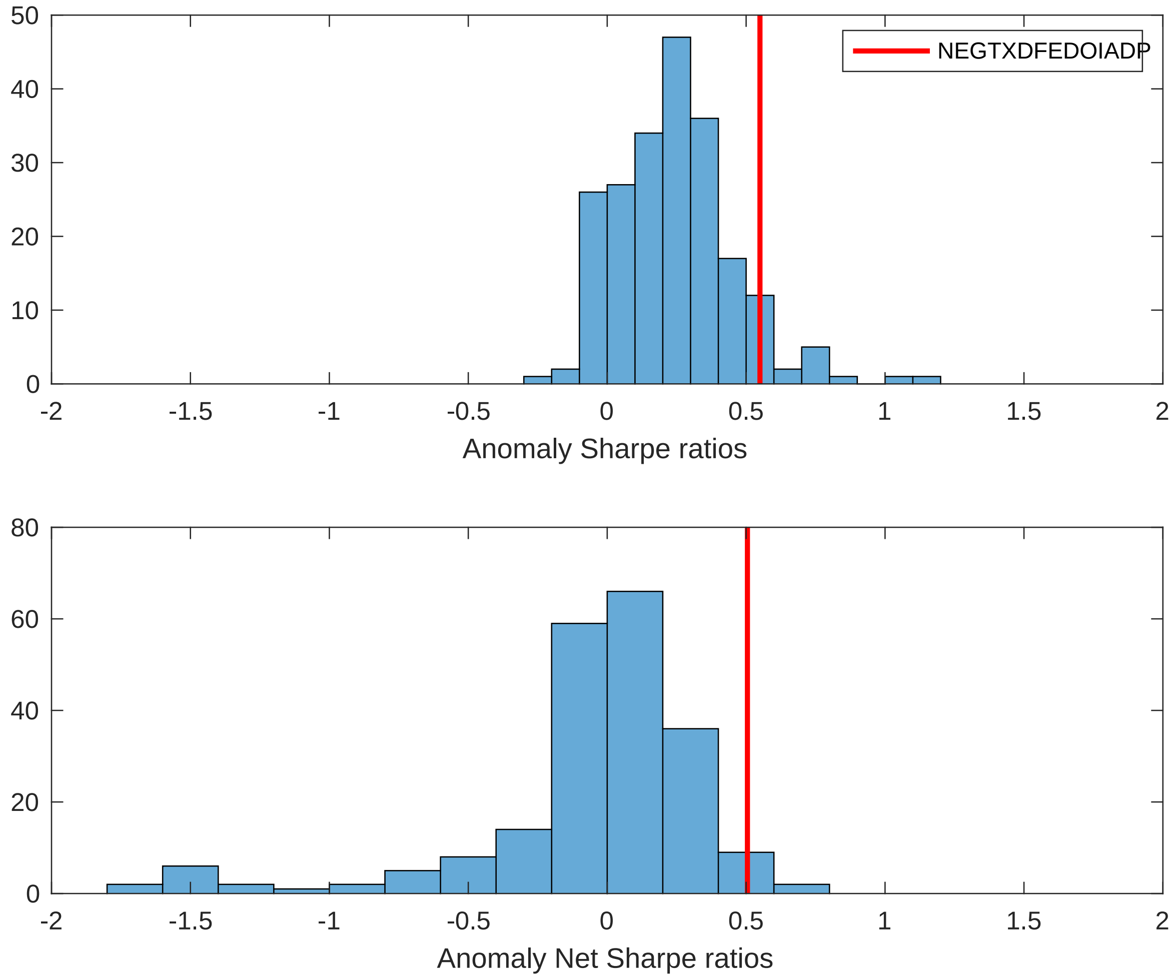


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

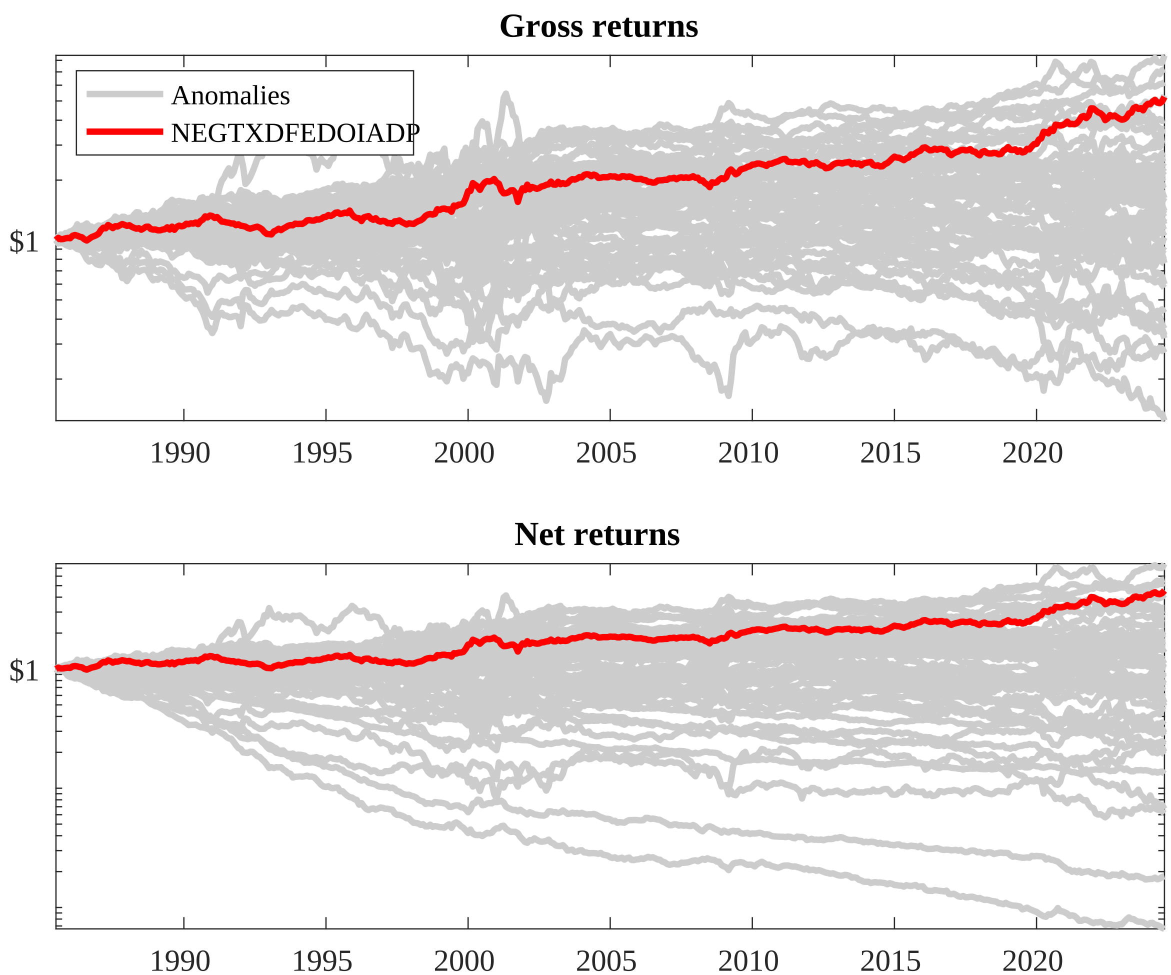
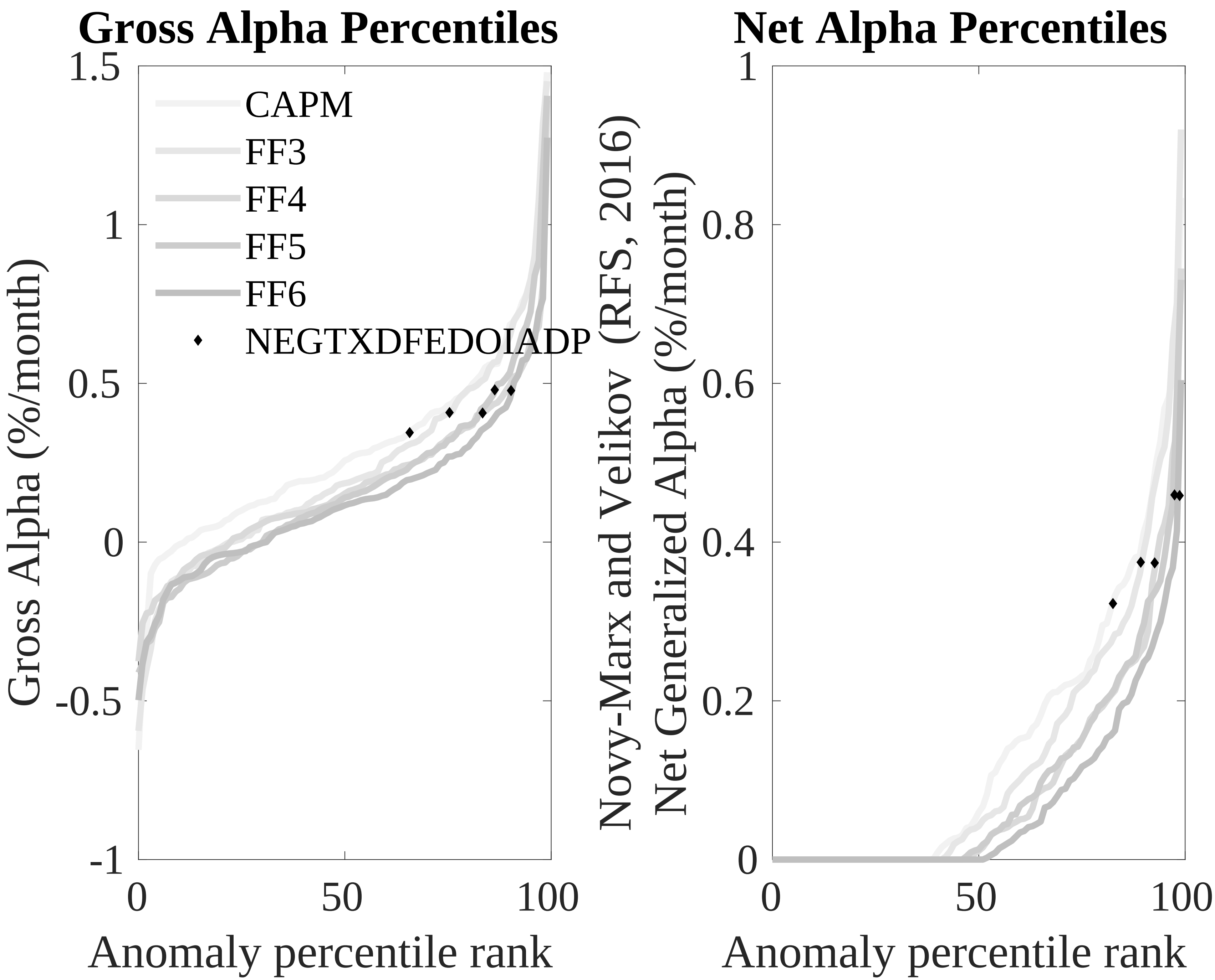
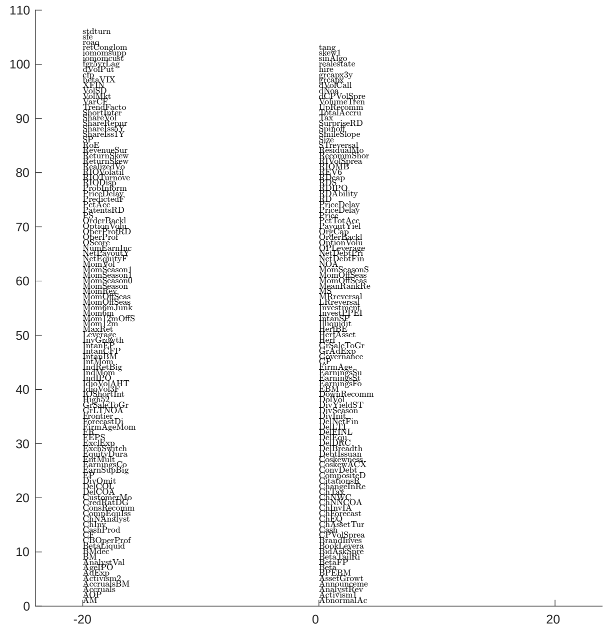


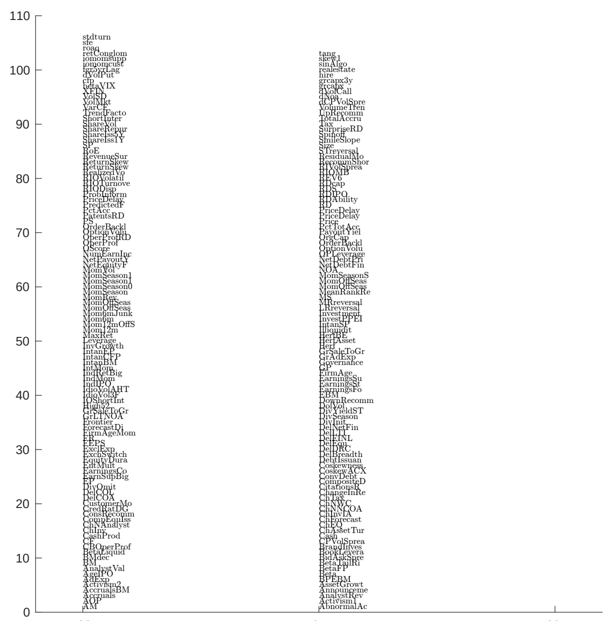
Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGTXDFEDOIADP
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
1
0.8
0.6
0.4
0.2
0
0
50
100
0
50
100



Panel A: Pearson



Panel B: Spearman

Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

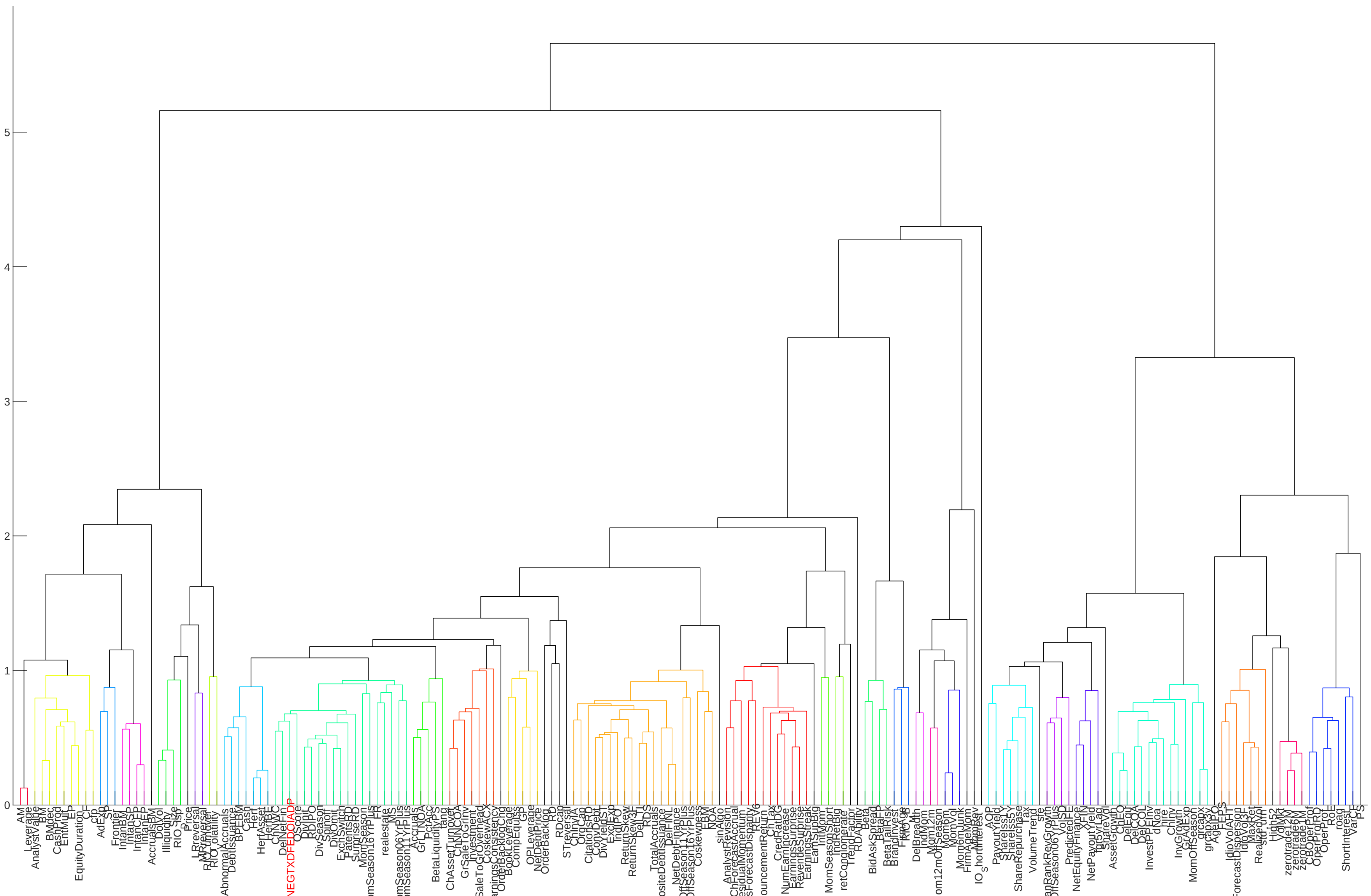


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

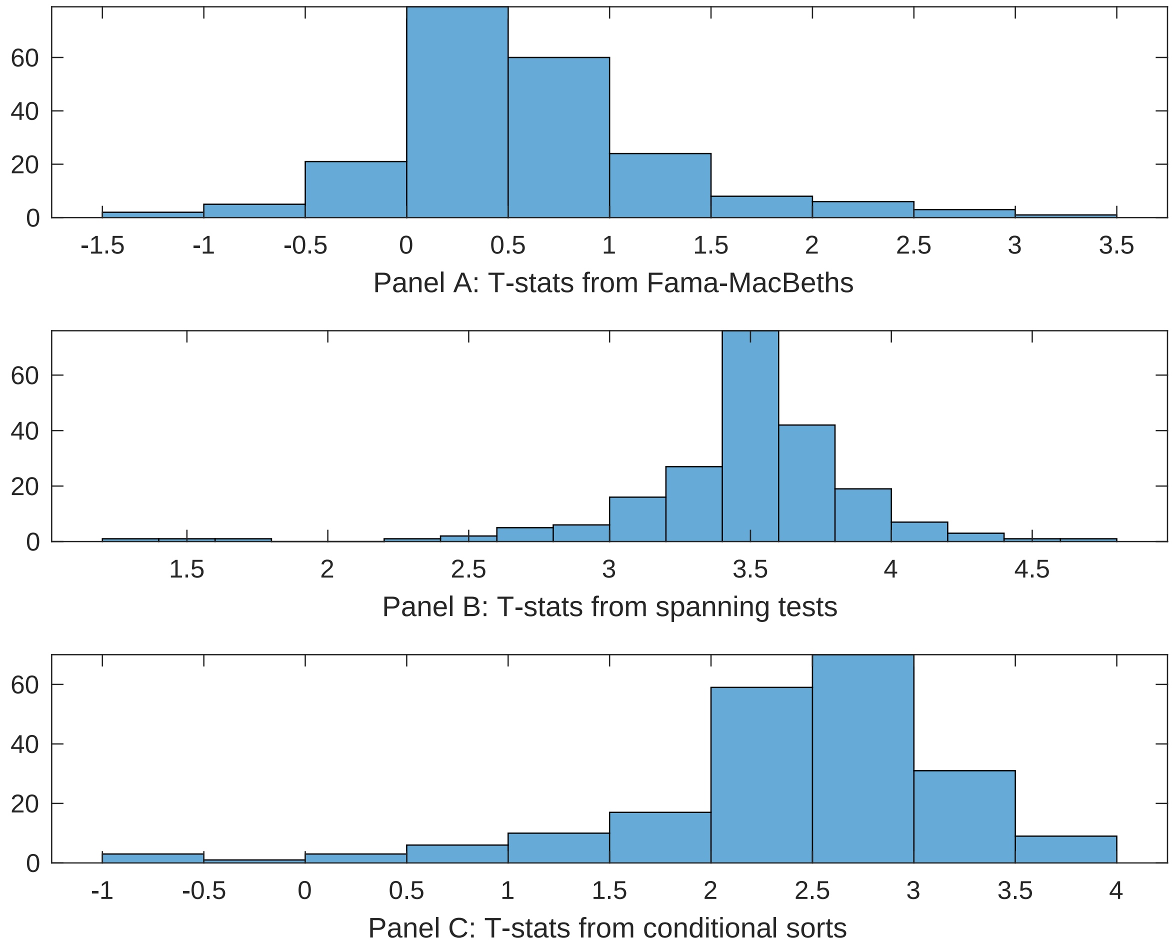


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.10	0.16	0.10	0.93	0.10	0.10	0.12
	[3.69]	[6.32]	[3.50]	[3.04]	[3.51]	[3.46]	[4.81]
TSI	0.78	0.18	0.24	0.34	0.39	0.34	0.43
	[1.35]	[0.31]	[0.43]	[0.59]	[0.69]	[0.59]	[0.77]
Anomaly 1	0.62						0.11
	[1.56]						[3.13]
Anomaly 2		0.25					0.23
		[2.66]					[3.41]
Anomaly 3			0.18				-0.23
			[0.58]				[-0.98]
Anomaly 4				0.27			0.20
				[3.22]			[2.84]
Anomaly 5					0.13		-0.25
					[0.61]		[-1.32]
Anomaly 6						0.85	0.79
						[2.95]	[4.36]
# months	463	468	463	463	463	463	463
$\bar{R}^2(\%)$	1	0	1	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Cash to assets, Equity Duration, Cash flow to market, Book to market using December ME, Market leverage, Operating Cash flows to price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198506 to 202406.

Intercept	0.28	0.47	0.48	0.46	0.47	0.51	0.29
	[2.84]	[4.63]	[4.73]	[4.56]	[4.56]	[5.00]	[2.90]
Anomaly 1	33.90						30.37
	[8.54]						[7.01]
Anomaly 2		-13.35					-1.87
		[-2.49]					[-0.29]
Anomaly 3			-8.48				2.14
			[-2.40]				[0.48]
Anomaly 4				-25.89			-9.98
				[-3.87]			[-1.45]
Anomaly 5					-12.58		-6.98
					[-3.00]		[-1.62]
Anomaly 6						-13.55	-3.78
						[-3.62]	[-0.87]
mkt	-6.58	-0.34	-0.21	-1.30	1.38	0.27	-5.33
	[-2.77]	[-0.14]	[-0.09]	[-0.54]	[0.56]	[0.11]	[-2.14]
smb	-6.87	-5.10	-7.16	-0.88	-7.09	-5.87	-3.35
	[-2.00]	[-1.33]	[-1.95]	[-0.22]	[-1.94]	[-1.60]	[-0.86]
hml	-12.23	-10.22	-19.96	1.74	-8.23	-14.92	9.45
	[-2.75]	[-1.36]	[-4.06]	[0.21]	[-1.16]	[-2.86]	[1.00]
rmw	-5.99	-22.11	-19.30	-29.19	-24.13	-17.73	-10.77
	[-1.26]	[-4.77]	[-4.01]	[-5.93]	[-5.18]	[-3.71]	[-1.97]
cma	18.37	1.37	8.59	6.95	3.20	8.92	15.48
	[2.92]	[0.20]	[1.29]	[1.07]	[0.49]	[1.36]	[2.28]
umd	0.42	0.48	-2.98	1.58	-2.06	-4.75	-1.50
	[0.20]	[0.21]	[-1.07]	[0.69]	[-0.83]	[-1.73]	[-0.49]
# months	464	468	464	464	464	464	464
$\bar{R}^2$(%)	29	19	18	20	19	19	29

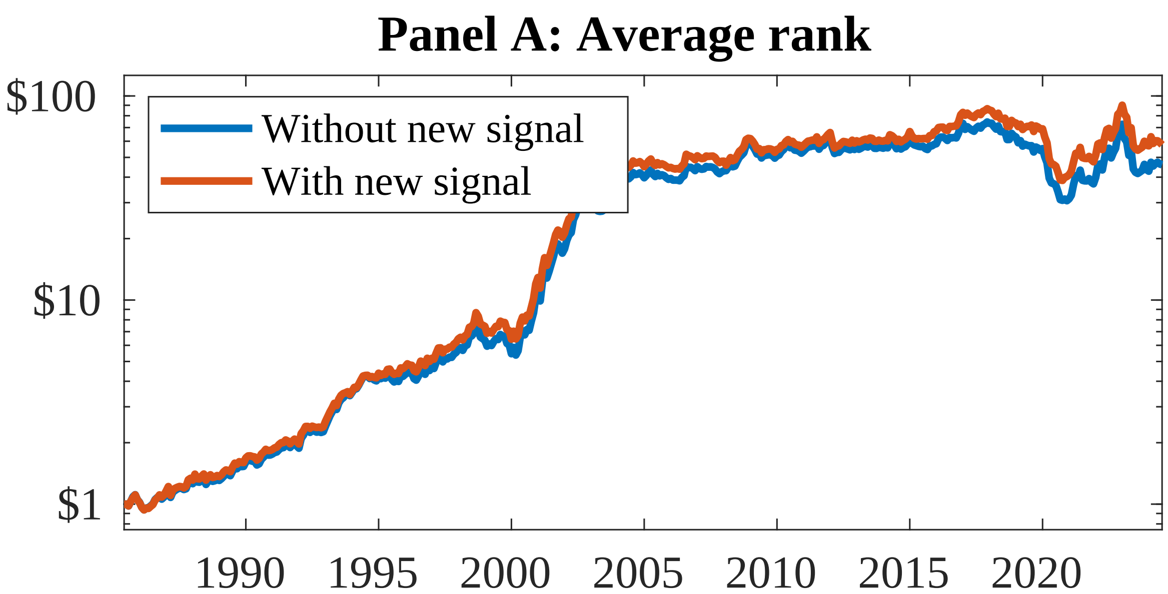


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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